



Md. Omar Sha Rafi

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RESEARCH INTEREST

My research focuses on natural products chemistry and phytochemistry, with particular emphasis on the isolation, characterization, and bioactivity evaluation of plant-derived bioactive compounds. I am interested in exploring multidisciplinary approaches that integrate phytochemical screening, analytical characterization, molecular docking, and in-vitro/in-vivo biological assays to identify and validate novel therapeutic agents from traditional medicinal plants. My work aims to bridge traditional herbal knowledge with modern drug discovery and pharmacological science, contributing to the development of biologically active molecules.

RESEARCH EXPERIENCE

- **Supervision:** Trained 6 MSc and 12 undergraduate students in compound isolation, organic synthesis, and analytical techniques.
- **M.S. Thesis:** Isolation of Phytoconstituents, Fatty Acid Analysis and Biological Investigation on the Flower of *Acmella calva* (DC.) R.K. Jansen (Surjo Konna)
 - Applied **spectroscopic and chromatographic methods** (NMR, UV-Vis, IR, XRD, HPLC) for structural elucidation and biological activity studies.
- **B.S. Project:** Phytochemical Screening, In-Vitro Cytotoxicity Evaluation and Fatty acid analysis of *Spilanthes acmella* L. Flower

PUBLICATIONS

Publications.

- Mimi, S. S., Fahim, M. S. Z., Rafi, M. O. S., Mia, M. S., & Hasan, M. K. (2026). Experimental and computational evaluation of bioactive secondary metabolites from *Trewia nudiflora* stem bark. *Fitoterapia*, 195, 107479.
- Al Mamun, A., Shanchary, H. R., *Rafi, M. O. S., Ray, S. K., Shandhi, S. P., Bhuiyan, R. H., Chowdhury, F., & Muslim, T. (2026). Innovative TiO₂-SiO₂-La₂O₃ nanocomposite systems for enhanced wastewater treatment performance. *Next Nanotechnology*, 10, 100807.
- Meem, M. I., Fahim, M. S. Z., Mia, M. S., *Rafi, M. O. S., & Akhter, K. (2026). Catechol substitution governs antioxidant potency in 5-arylidene-thiazolidine-2, 4-diones: Synthesis, structure-activity relationships, DFT, docking, and molecular dynamics. *Results in Chemistry*, 103793.
- Al Mamun, A., Ray, S. K., *Rafi, M. O. S., Shandhi, S. P., Bhuiyan, R. H., Chowdhury, F., & Muslim, T. (2026). Design, characterization, and application of a mixed metal oxide (Al₂O₃-La₂O₃-V₂O₅) nanocomposite in photocatalytic degradation of methylene blue: a multi-technique approach. *RSC advances*. (Scopus Indexed)
- Ahmed, M. S., *Rafi, M. O. S., & Muslim, T. (2026). Valorization of crab shell waste into chitin, chitosan, and carboxymethyl chitosan: Extraction, chemical modification, and characterization. *Next Materials*, 11, 101710. (Scopus Indexed)
- Fahim, M. S. Z., Moulick, S. P., Mimi, S. S., Dey, S. S., Rafi, M. O. S., Chhanda, S. A., ... & Muslim, T. (2026). Exploring the bioactivity of *Ageratum conyzoides* L.: Antioxidant, Antimicrobial, and Cytotoxic Properties of Flower Extracts. *Next Research*, 102216.
- *Rafi, M. O. S., Shristy, N. T., Mimi, S. S., Chhanda, S. A., Rahman, M. H., Rahman, M. S., ... & Muslim, T. (2025). Antioxidant, thrombolytic, cytotoxic and membrane stabilizing activities and fatty acid composition of *Acmella Calva* (DC.) Flower. *Bangladesh Journal of Botany*, 54(3), 433-439.
- Hossain, R. B., Mimi, S. S., *Rafi, M. O. S., Chhanda, S. A., Ibrahim, M., & Muslim, T. (2026). In vivo Evaluation of Analgesic, Antidiarrheal, Hypoglycemic, and Anti-Inflammatory Activities of *Sarcolobus*

carinatus Methanolic Extract in Swiss Albino Mice. *Bangladesh Pharmaceutical Journal*, 29(2), 282-289.

- Bristy, R. R., Mimi, S. S., Chhanda, S. A., Ali, M. H., Rafi, M. O. S., & Muslim, T. (2025). Investigation of the Pharmacological Potential of *Staurogyne zeylanica* Kuntze: In vitro and In vivo Studies. *Bangladesh Pharmaceutical Journal*, 28(1), 28-36.
- Jahan, I., Ali, M. H., Shristy, N. T., Rafi, M. O. S., Mimi, S. S., Siddik, M. N. A., ... & Chowdhury, T. A. (2024). Assessing Therapeutic Potentials of *Catharanthus roseus* (L.) G. Don Focusing on Anti-Diabetic, Analgesic and Anti-Diarrheal Activities. *Bangladesh Pharmaceutical Journal*, 27(2), 215-222.
- Shristy, N. T., Rafi, M. O. S., Mimi, S. S., Chowdhury, V., Rahman, M. H., & Chowdhury, T. A. (2024). In vitro Biological Activities and Fatty Acid Composition Analysis of *Musa paradisiaca* L. Inflorescence. *Dhaka University Journal of Pharmaceutical Sciences*, 23(1), 69-76.
- Rafi, M. O. S., Shristy, N. T., Chowdhury, V., Rahman, M. H., Mimi, S. S., Afroz, R., & Chowdhury, T. A. (2023). *Acmella calva* (DC.) R.K. Jansen flower: A comprehensive in vivo study of its central and peripheral analgesic, hypoglycemic, antidepressant, and anti-diarrheal properties. *Dhaka University Journal of Pharmaceutical Sciences*, 22(1), 65-71.

Conference Presentations

- Rafi, Md. Omar Sha; Rahman, Md. Hasanur; Chowdhury, Tofail Ahmad; “Isolation of Phytoconstituents and Biological Investigation on the Flower of *Acmella calva* (DC.) R.K. Jansen (Surjo Konna)” *BCSIR Congress on Integrated Approach for Adapting 4IR* December 1-3, 2022, Bangladesh Council of Scientific and Industrial Research (BCSIR), Dhaka, Bangladesh (Oral Presentation)
- Shristy, Nusrat Tabassum; Rafi, Md. Omar Sha; Chowdhury, Vaskar; Rahman, Md. Hasanur; Chowdhury, Tofail Ahmad; “Phytochemical and pharmacological investigations of the inflorescences of *Musa paradisiaca* L. (*Atia kola*)” *BCSIR Congress on Integrated Approach for Adapting 4IR* December 1-3, 2022, Bangladesh Council of Scientific and Industrial Research (BCSIR), Dhaka, Bangladesh (Oral Presentation).

WORK EXPERIENCE

Lecturer of Chemistry

Department of Chemistry, University of Dhaka

Dec 2023–Present

- **Led 3 externally funded research projects (PI/Co-I), securing ~1 million BDT**, producing 3 publications.
- **Supervised 6 MSc and 12 undergraduate students** in compound isolation, organic synthesis, and advanced analytical techniques.
- **Authored/co-authored 12 peer-reviewed publications.**
- **Designed and taught 2+ theory courses and 4 lab courses** in Organic and General Chemistry, enhancing student learning outcomes.

Research Fellow

Bose Center for Advanced Study and Research in Natural Sciences, University of Dhaka

01/07/2023 – 01/12/2023

Project Title: Evaluation of Phytochemicals and Bioactive Properties of *Adensoma indianum*.

Research Assistant

Centennial Research Grant (CRG 2nd Phase), University of Dhaka

01/07/2022 – 30/06/2023

Project Title: Investigation of Nutritional Content, Phytochemical and Biological Activity on the Fruit of *Limonia acidissima* L.

EDUCATION

MS in Organic Chemistry, University of Dhaka | CGPA 3.91/4.00

Aug 2021 - Dec 2022

SKILLS

- Software: MS-Word, MS-Excel, MS PowerPoint, SigmaPlot, ChemDraw, PyRx, Discovery Studio, Autodock Vina
- Instrumental: FT-IR, UV Spectrophotometer, GC-FID
- Language: Bangla, English
- Programming Skill: Python 3, Javascript, HTML, Ruby, Bash
- Others: Excellent presentation and communication skills, Good teaching skills, Team work, Critical thinking, Good organization and management skills, Analytical and problem-solving skills, Synack red team member

RESEARCH AND GRANTS

- **Principal Investigator:** *Isolation of Secondary Metabolites, Fatty Acid Analysis and Biological Investigation on the Flower of *Acmella calva* (DC.) R.K. Jansen (Surjo Konna)*, Funded by Ministry of Science and Technology (MOST), Bangladesh, 80,000 BDT. Experimental study exploring therapeutic potential.

Outcomes: Resulted in 1 peer-reviewed publication and the training of 2 MSc students.

- **Principal Investigator:** *Phytochemical and Pharmacological Exploration of *Trianthema portulacastrum* L. Leaves: A Potential Medicinal Weed*, Funded by Ministry of Science and Technology (MOST), Bangladesh, 80,000 BDT. Experimental study exploring therapeutic potential.
- **Co-Investigator:** *Synthesis and Structure-Activity Relationships of Thiazolidinedione Derivatives as Potential Therapeutic Agents*—Funded by University Grants Commission (UGC), Bangladesh, 750,000 BDT. Contributed to project design, experimentation, and data analysis.

HONORS & AWARDS

- National Science and Technology (NST) fellowship (December, 2021)
- University Merit Scholarship Based on academic excellence at B. S (April, 2022)
- Mitsubishi Corporation International Scholarship (2018)
- Dhaka Board Government Merit Scholarship award based on academic excellence at Secondary School Certificate (2014)
- Dhaka Board Government Merit Scholarship award based on academic excellence at Junior School Certificate (2010)
- Dhaka Board Government Merit Scholarship award based on academic excellence at Primary Scholarship exam (2007)
- 1st runner up at HackerOne BUG HUNT 2023 organised by bug bounty community Bangladesh at ICT Division, Agargaon, Dhaka